Gemini

data_carseats <- Carseats
summary(data_carseats)

```
> summary(data_carseats)
     Sales          CompPrice       Income        Advertising       Population
 Min.   : 0.000   Min.   : 77   Min.   : 21.00   Min.   : 0.000   Min.   : 10.0
 1st Qu.: 5.390   1st Qu.:115   1st Qu.: 42.75   1st Qu.: 0.000   1st Qu.:139.0
 Median : 7.490   Median :125   Median : 69.00   Median : 5.000   Median :272.0
 Mean   : 7.496   Mean   :125   Mean   : 68.66   Mean   : 6.635   Mean   :264.8
 3rd Qu.: 9.320   3rd Qu.:135   3rd Qu.: 91.00   3rd Qu.:12.000   3rd Qu.:398.5
 Max.   :16.270   Max.   :175   Max.   :120.00   Max.   :29.000   Max.   :509.0
     Price        ShelveLoc        Age          Education    Urban       US
 Min.   : 24.0   Bad   : 96   Min.   :25.00   Min.   :10.0   No :118   No :142
 1st Qu.:100.0   Good  : 85   1st Qu.:39.75   1st Qu.:12.0   Yes:282   Yes:258
 Median :117.0   Medium:219   Median :54.50   Median :14.0
 Mean   :115.8                Mean   :53.32   Mean   :13.9
 3rd Qu.:131.0                3rd Qu.:66.00   3rd Qu.:16.0
 Max.   :191.0                Max.   :80.00   Max.   :18.0
```

Identifying Skewness and Distribution

In your previous exercise, you looked for a large gap between the **Mean** and **Median** to identify skewness.

- **Sales, CompPrice, and Price**: These appear very symmetric. For example, `Sales` has a Median of **7.490** and a Mean of **7.496**. You likely won't need a `log()` transformation for these.
- **Advertising**: This is the most skewed variable. The Mean (**6.635**) is higher than the Median (**5.000**). More importantly, the **1st Quartile is 0.000**, confirming that at least 25% of the stores have zero advertising spend.

2. Categorical Variables (Factors)

Unlike the numeric variables, R summarizes these by counting the occurrences:

- **ShelveLoc**: Has three levels (Bad: 96, Good: 85, Medium: 219). When you fit your regression, you should use `as.factor(ShelveLoc)` or ensure R treats it as a factor to create dummy variables.
- **Urban & US**: These are binary (Yes/No).

```
Residuals:
    Min      1Q  Median      3Q     Max 
-2.8799 -0.7015  0.0088  0.6611  3.4268 

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)    
(Intercept)                 5.761899   0.575319  10.015  < 2e-16 ***
Price                      -0.095302   0.002667 -35.736  < 2e-16 ***
CompPrice                   0.092609   0.004128  22.436  < 2e-16 ***
Income                      0.015774   0.001843   8.560 2.65e-16 ***
Advertising                 0.125044   0.010558  11.844  < 2e-16 ***
as.factor(ShelveLoc)Good    4.846736   0.152852  31.709  < 2e-16 ***
as.factor(ShelveLoc)Medium  1.952145   0.125732  15.526  < 2e-16 ***
Age                        -0.046119   0.003176 -14.520  < 2e-16 ***
Education                  -0.022411   0.019565  -1.145    0.253    
UrbanYes                    0.118853   0.112648   1.055    0.292    
USYes                      -0.199075   0.147315  -1.351    0.177    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.018 on 389 degrees of freedom
Multiple R-squared:  0.8733,	Adjusted R-squared:  0.8701 
F-statistic: 268.2 on 10 and 389 DF,  p-value: < 2.2e-16
```

Variable	Estimate	Significance (p-value)	Interpretation
Price	−0.095	<2×10−16	For every $1 increase in Price, Sales decrease by 0.095 units.
Advertising	0.125	<2×10−16	Every additional unit of Advertising budget increases Sales by 0.125 units.
ShelveLocGood	4.846	<2×10−16	Having a "Good" shelf location increases Sales by 4.84 units compared to the "Bad" baseline.
Age	−0.046	<2×10−16	As the average age of the local population increases by one year, Sales decrease by 0.046 units.
Education	−0.022	0.253	This variable is **not statistically significant** (p>0.05), meaning we do not have enough evidence to say it affects Sales.

Based on the R output from the `Carseats` dataset analysis, the linear regression model can be expressed in the following equation form:

$$= \beta_0 + \beta_1(i) + \beta_2(i) + \beta_3(n) + \beta_4(iin) + \ + \epsilon$$

Using the specific **Estimate** values from your screenshot, the equation for the predicted sales is:

$$= 5.71 - 0.053(i) + 0.02(i) + 0.015(n) + 0.1250(iin) + 4.47(L) + 1.521(Li) - 0.041() - 0.0224(in) +$$

Key Components of the Equation

- **The Intercept ($\beta_0 = 5.71$):** This represents the predicted sales value if all other independent variables were zero.
- **The Coefficients (β_j):** Each value (e.g., -0.053 for Price) represents the expected change in Sales for a one-unit increase in that variable, assuming all other variables stay constant.
- **Dummy Variables:** Notice that categorical variables like `ShelveLoc` are broken into separate terms (Good and Medium). This is because R automatically creates **dummy variables** for factors to include them in the **Design Matrix** (X).
- **The Matrix Context:** In the background, R calculates these specific numbers using the general matrix formula $Y = X +$, where the vector of estimates $\hat{}$ is found by solving $\hat{\beta} = (XX)^{-1}XY$.
  Based on the R regression output provided, you can reject the null hypothesis $H_0 : \beta_j = 0$ for any predictor where the **p-value** (listed in the `Pr(>|t|)` column) is less than your significance level, typically $= 0.05$.

You can reject the null hypothesis for the following predictors:

- **Price**: The p-value is $2 - 1$, which is well below 0.05.
- **CompPrice**: The p-value is $2 - 1$.
- **Income**: The p-value is $2.5 - 1$.
- **Advertising**: The p-value is $2 - 1$.
- **as.factor(ShelveLoc)Good**: The p-value is $2 - 1$.
- **as.factor(ShelveLoc)Medium**: The p-value is $2 - 1$.
- **Age**: The p-value is $2 - 1$.

Why These?

In the general model, the p-value represents the probability of observing the calculated test statistic if the null hypothesis H_0 were true. Since these p-values are extremely small, we conclude that these variables have a statistically significant relationship with **Sales** and are unlikely to have a coefficient of zero.

Which cannot be rejected?

You **cannot** reject the null hypothesis for **Education** ($= 0.253$), **UrbanYes** ($= 0.22$), or **USYes** ($= 0.177$), as their p-values are all greater than 0.05. This suggests that, in this specific model,

we do not have enough evidence to say these factors influence Sales.

```
Residuals:
    Min      1Q  Median      3Q     Max 
-2.7728 -0.6954  0.0282  0.6732  3.3292 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)      5.475226   0.505005   10.84   <2e-16 ***
Price           -0.095319   0.002670  -35.70   <2e-16 ***
CompPrice        0.092571   0.004123   22.45   <2e-16 ***
Income           0.015785   0.001838    8.59   <2e-16 ***
Advertising      0.115903   0.007724   15.01   <2e-16 ***
ShelveLocGood    4.835675   0.152499   31.71   <2e-16 ***
ShelveLocMedium  1.951993   0.125375   15.57   <2e-16 ***
Age             -0.046128   0.003177  -14.52   <2e-16 ***
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.019 on 392 degrees of freedom
Multiple R-squared:  0.872,	Adjusted R-squared:  0.8697 
F-statistic: 381.4 on 7 and 392 DF,  p-value: < 2.2e-16
```

Comparison of Fit Metrics

Based on the summaries for the full model and the smaller model, here is how they compare:

Metric	Full Model (a)	Smaller Model (e)	Better Fit?
Residual Standard Error (RSE)	1.018	1.019	**Full Model (slightly)**
Multiple R-squared	0.8733	0.872	**Full Model**
Adjusted R-squared	0.8701	0.8697	**Full Model (slightly)**
F-statistic	268.2	381.4	**Smaller Model**

Analysis of the Variables

- **Adjusted R-squared**: This variable is used to distinguish fit while penalizing for unnecessary predictors. In this case, the Adjusted R-squared dropped very slightly (from 0.8701 to 0.8697) when the non-significant variables were removed. This suggests that **Education**, **Urban**, and **US** contributed almost no meaningful explanatory power to the model.
- **F-statistic**: The F-statistic increased significantly from 268.2 to 381.4. A higher F-statistic indicates that the smaller model is more "parsimonious," meaning it provides a stronger

overall test of the remaining significant predictors.

- **Residual Standard Error (RSE)**: The error remains nearly identical (1.018 vs 1.019). This shows that the average distance of the data points from the regression line did not meaningfully change despite removing three variables.

Conclusion on Fit

While the **Full Model** has a mathematically "better" fit by a negligible margin in terms of Adjusted R-squared, the **Smaller Model** is considered superior for practical use. It achieves nearly the same predictive performance while being simpler and ensuring all included variables are statistically significant.

(g) 95% Confidence Intervals for the Coefficients

To calculate the 95% confidence interval for any coefficient β_j, you use the estimate and its standard error from the R output. The general formula is:

$$\hat{\beta}_j \ (\ \ (\hat{\beta}_j))$$

For a large sample size like this ($n = 400,\ f = 32$), the critical value is approximately **1.96**. Based on the "Smaller Model" screenshot:

- **Price:** $-0.053\ \ (1.\ \ 0.0027)$ **[-0.1005, -0.0901]**
- **Advertising:** $0.115\ \ (1.\ \ 0.00772)$ **[0.1008, 0.1310]**
- **Income:** $0.015\ \ (1.\ \ 0.0014)$ **[0.0122, 0.0194]**

> **Interpretation:** We are 95% confident that for every $1 increase in Price, Sales will decrease by between 0.09 and 0.10 units, holding other factors constant.

(h) Evidence of Outliers or High Leverage Observations

To determine if specific data points are "breaking" the optimality of your **Least Squares Estimator**, we look at the residuals and the range of predictors.

1. Outliers (Residual Analysis)

- **Observation:** The residuals in Model (e) range from **-2.77 to 3.33**.
- **Evidence:** Standardized residuals greater than $|3|$ are typically considered potential outliers. Since the maximum residual is **3.3292**, there is at least one observation that the model predicts poorly (an outlier).

2. High Leverage Observations

- **Definition:** Leverage measures how far an individual's predictor values (like a massive `Income` or extreme `Price`) are from the average of the other observations.
- **Evidence:** In your preparatory screencast, the instructor noted that `Income` reached as high as **10 million** and `Face` value reached **14 million**. Such extreme values (even after logging) often result in high leverage, meaning those specific data points have a disproportionately large influence on the "tilt" of the regression line.
- **Risk:** As noted in your text on **extrapolation**, if these high-leverage points fall outside the typical range of the covariates, the estimated values can be misleading.

Answer log

Q10

(a) Fit a multiple regression model to predict Sales using Price, Urban, and US.

```
?Carseats
```

Dataset: Carseats (Sales of Child Car Seats)

This is a simulated dataset containing sales data for child car seats across **400 different stores**. It is frequently used to demonstrate multiple linear regression and the handling of qualitative (categorical) variables.

Variable Descriptions

Variable	Description
Sales	Unit sales (in thousands) at each location
CompPrice	Price charged by competitor at each location
Income	Community income level (in thousands of dollars)
Advertising	Local advertising budget for company (in thousands of dollars)
Population	Population size in region (in thousands)
Price	Price the company charges for car seats at each site
Age	Average age of the local population
Education	Education level at each location

Qualitative (Categorical) Variables

These variables are factors in R and require special handling (dummy coding) during regression analysis:

- **ShelveLoc**: Quality of the shelving location at the site.
  - *Levels:* **Bad**, **Medium**, **Good**
- **Urban**: Indicates if the store is in an urban or rural location.
  - *Levels:* **No**, **Yes**
- **US**: Indicates if the store is located in the US.
  - *Levels:* **No**, **Yes**

```
summary(Carseats)
```

Data cleaning decisions:

```
> summary(Carseats)
     Sales          CompPrice       Income        Advertising
 Min.   : 0.000   Min.   : 77   Min.   : 21.00   Min.   : 0.000
 1st Qu.: 5.390   1st Qu.:115   1st Qu.: 42.75   1st Qu.: 0.000
 Median : 7.490   Median :125   Median : 69.00   Median : 5.000
 Mean   : 7.496   Mean   :125   Mean   : 68.66   Mean   : 6.635
 3rd Qu.: 9.320   3rd Qu.:135   3rd Qu.: 91.00   3rd Qu.:12.000
 Max.   :16.270   Max.   :175   Max.   :120.00   Max.   :29.000
   Population        Price        ShelveLoc        Age
 Min.   : 10.0   Min.   : 24.0   Bad   : 96   Min.   :25.00
 1st Qu.:139.0   1st Qu.:100.0   Good  : 85   1st Qu.:39.75
 Median :272.0   Median :117.0   Medium:219   Median :54.50
 Mean   :264.8   Mean   :115.8                Mean   :53.32
 3rd Qu.:398.5   3rd Qu.:131.0                3rd Qu.:66.00
 Max.   :509.0   Max.   :191.0                Max.   :80.00
   Education    Urban       US
 Min.   :10.0   No :118   No :142
 1st Qu.:12.0   Yes:282   Yes:258
 Median :14.0
 Mean   :13.9
 3rd Qu.:16.0
 Max.   :18.0
```

- No NAs to consider in any variables
- Boundaries (Max & Min of all variables make sense) except Education which isnt completely clear?
- Correlation between variables

```
cor(Carseats$Price, Carseats$Sales)
[1] -0.4449507
# Expected negative correlation between the two
```

Fitting the model

```
lm.fit <- lm(Sales ~ Price + Urban + US, data = Carseats)
summary(lm.fit)
```

```
Call:
lm(formula = Sales ~ Price + Urban + US, data = Carseats)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.9206 -1.6220 -0.0564  1.5786  7.0581 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 13.043469   0.651012  20.036  < 2e-16 ***
Price       -0.054459   0.005242 -10.389  < 2e-16 ***
UrbanYes    -0.021916   0.271650  -0.081    0.936    
USYes        1.200573   0.259042   4.635 4.86e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.472 on 396 degrees of freedom
Multiple R-squared:  0.2393,	Adjusted R-squared:  0.2335 
F-statistic: 41.52 on 3 and 396 DF,  p-value: < 2.2e-16
```

(b) Provide an interpretation of each coefficient in the model. Be careful—some of the variables in the model are qualitative!

The summary results indicate that a unit increase in sales means that price decreased by 0.054459 units. The UrbanYes variable is categorical, the estimated negative negative coefficient (-0.021916) indicates sales are lower in urban-located stores. This is the opposite for US-based stores with 1.2 more units.

It is noted that the categorical variables are the 'yes' options. How does this work?

(c) Write out the model in equation form, being careful to handle the qualitative variables properly.

$$y() = -0.05445i + -0.0211n + 1.200573 + 13.0434$$

(d) For which of the predictors can you reject the null hypothesis H0 : βj = 0?

R makes it even easier to scan these values by adding "Significance Codes" (stars) to the right of the p-value:

- `*`**: Very highly significant (p<0.001).
- ``**: Highly significant (p<0.01).
- `*`: Significant (p<0.05).
- **(Empty)**: Not significant (p>0.1)

`Pr(>|t|)` is the **p-value**
Price: < 2e-16 ***

UrbanYes: 0.936 (Empty), So the relationship between urban or rural locations is not significant in its relationship with sales.

USYes: 4.86e-06 ***

(e) On the basis of your response to the previous question, fit a smaller model that only uses the predictors for which there is evidence of association with the outcome.

```
lm.fit_small <- lm(Sales ~ Price + US, data = Carseats)
summary(lm.fit_small)
```

```
Call:
lm(formula = Sales ~ Price + US, data = Carseats)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.9269 -1.6286 -0.0574  1.5766  7.0515 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 13.03079    0.63098  20.652  < 2e-16 ***
Price       -0.05448    0.00523 -10.416  < 2e-16 ***
USYes        1.19964    0.25846   4.641 4.71e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.469 on 397 degrees of freedom
Multiple R-squared:  0.2393,	Adjusted R-squared:  0.2354 
F-statistic: 62.43 on 2 and 397 DF,  p-value: < 2.2e-16
```

(f) How well do the models in (a) and (e) fit the data?

Comparing the "Readings"

Metric	Model (a): Price + Urban + US	Model (e): Price + US
Residual Standard Error (RSE)	2.472	2.469
Multiple R2	0.2393	0.2393
Adjusted R2	0.2335	0.2354

- **RSE (Accuracy of Predictions):** The RSE measures the average amount that the response (`Sales`) will deviate from the regression line. In the smaller model (e), the RSE is slightly lower (2.469 vs 2.472). This means model (e) is marginally more precise.
- **R2 (Variance Explained):** Both models explain approximately **23.9%** of the variance in sales. While this isn't a very "high" R2, it is typical for real-world sales data where many other factors (like the quality of the product or local competition) aren't in the model yet.
- **Adjusted R2 (Model Efficiency):** This is the most important reading here. Because the
- **Adjusted R2 increased**(from 0.2335 to 0.2354) when we removed `Urban`, it confirms that `Urban` was "dead weight". It was adding complexity without adding any actual predictive power.

Overall, the smaller RSE and greater adjusted R^2 value means the smaller model fits better and is parsimonious, achieving a better fit with less variables.

(g) Using the model from (e), obtain 95 % confidence intervals for the coefficient(s).

```
confint(lm.fit_small)
```

```
> confint(lm.fit_small)
                  2.5 %      97.5 %
(Intercept) 11.79032020 14.27126531
Price       -0.06475984 -0.04419543
USYes        0.69151957  1.70776632
```

- **Price:** The interval is **[−0.0647,−0.0441]**.
  - Since the entire range is **negative** and **does not include zero**, we are 95% confident that increasing the Price always results in a decrease in Sales.
- **USYes:** The interval is **[0.6915,1.7077]**.
  - Since the entire range is **positive** and **does not include zero**, we are 95% confident that being in the US always results in higher average Sales compared to non-US locations.

(h) Is there evidence of outliers or high leverage observations in the model from (e)?

```
png("my_plot.png", width = 800, height = 600)
plot(predict(lm.fit_small), rstudent (lm.fit_small))
dev.off()
```

Q15

(a) For each predictor, fit a simple linear regression model to predict the response. Identify which models show a statistically significant association and provide plots to support your assertions.

Boston Housing Dataset Overview

A dataset containing housing values and neighborhood characteristics for **506 suburbs** of Boston.

Variable Dictionary

Category	Variable	Description
Target	`medv`	**Median value** of owner-occupied homes (in $1,000s)
Demographics	`crim`	Per capita **crime rate** by town
	`zn`	Proportion of residential land zoned for lots > 25,000 sq.ft.
	`lstat`	Percentage of **lower status** population
	`ptratio`	**Pupil-teacher ratio** by town
Infrastructure	`rad`	Index of accessibility to **radial highways**
	`tax`	Full-value **property-tax rate** per $10,000
	`indus`	Proportion of **non-retail business** acres per town
Geography	`chas`	**Charles River** dummy variable (1 if tract bounds river; 0 otherwise)
	`dis`	Weighted mean of **distances** to 5 Boston employment centres
Environment	`nox`	**Nitrogen oxides** concentration (parts per 10 million)
	`rm`	Average number of **rooms** per dwelling

Category	Variable	Description
	`age`	Proportion of owner-occupied units **built prior to 1940**

Predictor	Coefficient (Estimate)	P-Value	Significance
crim	-0.41519	< 2e-16	Highly Significant
zn	0.14214	< 2e-16	Highly Significant
indus	-0.64849	< 2e-16	Highly Significant
chas	6.3462	7.39e-05	Highly Significant
nox	-33.916	< 2e-16	Highly Significant
rm	9.102	< 2e-16	Highly Significant
age	-0.12316	< 2e-16	Highly Significant
dis	1.0916	1.21e-08	Highly Significant
rad	-0.40310	< 2e-16	Highly Significant
tax	-0.025568	< 2e-16	Highly Significant
ptratio	-2.157	< 2e-16	Highly Significant
lstat	-0.95005	< 2e-16	Highly Significant

```
Boston
head(Boston)

### Fitting the model
lm.fit_crim <- lm(medv ~ crim, data = Boston)
lm.fit_zn<- lm(medv ~ zn, data = Boston)
lm.fit_lstat <- lm(medv ~ lstat, data = Boston)
lm.fit_ptratio <- lm(medv ~ ptratio, data = Boston)
lm.fit_rad <- lm(medv ~ rad, data = Boston)
lm.fit_tax <- lm(medv ~ tax, data = Boston)
lm.fit_indus <- lm(medv ~ indus, data = Boston)
lm.fit_chas <- lm(medv ~ chas, data = Boston)
lm.fit_dis <- lm(medv ~ dis, data = Boston)
lm.fit_nox <- lm(medv ~ nox, data = Boston)
lm.fit_rm <- lm(medv ~ rm, data = Boston)
lm.fit_age <- lm(medv ~ age, data = Boston)

summary(lm.fit_crim)
summary(lm.fit_zn)
summary(lm.fit_indus)
```

```
summary(lm.fit_chas)
summary(lm.fit_nox)
summary(lm.fit_rm)
summary(lm.fit_age)
summary(lm.fit_dis)
summary(lm.fit_rad)
summary(lm.fit_tax)
summary(lm.fit_ptratio)
summary(lm.fit_lstat)

# 1. Setup the grid (3 rows, 4 columns)
par(mfrow = c(3, 4))

# 2. List the variable names exactly as they appear in the dataset
# (Excluding 'medv' because it's our Y-axis)
predictors <- c("crim", "zn", "indus", "chas", "nox", "rm",
                "age", "dis", "rad", "tax", "ptratio", "lstat")

# 3. The Loop
for (var in predictors) {
  # Plot the raw data
  plot(Boston[[var]], Boston$medv,
       main = var,
       xlab = var,
       ylab = "medv",
       pch = 20,
       col = "darkgrey")

  # Construct the name of your model (e.g., "lm.fit_crim")
  # and use get() to fetch the actual model object
  model_name <- paste0("lm.fit_", var)
  model_obj <- get(model_name)

  # Draw the regression line from that specific model
  abline(model_obj, col = "red", lwd = 2)
}
```

(b) Fit a multiple regression model using all predictors. Identify for which predictors you can reject the null hypothesis $H_0 : \beta_j = 0$.

```
lm.fit_Boston <- lm(medv ~ ., data = Boston)
```

```
summary(lm.fit_Boston)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  41.617270   4.936039   8.431 3.79e-16 ***
crim         -0.121389   0.033000  -3.678 0.000261 ***
zn            0.046963   0.013879   3.384 0.000772 ***
indus         0.013468   0.062145   0.217 0.828520
chas          2.839993   0.870007   3.264 0.001173 **
nox         -18.758022   3.851355  -4.870 1.50e-06 ***
rm            3.658119   0.420246   8.705  < 2e-16 ***
age           0.003611   0.013329   0.271 0.786595
dis          -1.490754   0.201623  -7.394 6.17e-13 ***
rad           0.289405   0.066908   4.325 1.84e-05 ***
tax          -0.012682   0.003801  -3.337 0.000912 ***
ptratio      -0.937533   0.132206  -7.091 4.63e-12 ***
lstat        -0.552019   0.050659 -10.897  < 2e-16 ***
---
```

indus, age

(c) Compare the results from (a) and (b). Create a plot with the univariate (simple) regression coefficients on the x-axis and the multiple regression coefficients on the y-axis to visualize how the estimates change.

Plot the two different estimated coefficients on either axes. (Simple on x)(Multiple on y)

Simple regression:
crim: -0.41519
zn: 0.14214
indus: -0.64849
chas 6.3462
nox -33.916
rm 9.102
age -0.12316
dis 1.0916
rad -0.40310
tax -0.025568
ptratio -2.157
lstat -0.95005

Multiple regression model:

Predictor	Coefficient (Estimate)	P-Value	Interpretation
crim	-0.121389	0.000261	Significant negative impact
zn	0.046963	0.000772	Significant positive impact
indus	0.013468	0.828520	**Not Significant**
chas	2.839993	0.001173	Significant positive impact
nox	-18.758022	1.50e-06	Significant negative impact
rm	3.658119	< 2e-16	Strong significant positive impact
age	0.003611	0.786595	**Not Significant**
dis	-1.490754	6.17e-13	Significant negative impact
rad	0.289405	1.84e-05	Significant positive impact
tax	-0.012682	0.000912	Significant negative impact
ptratio	-0.937533	4.63e-12	Significant negative impact
lstat	-0.552019	< 2e-16	Strong significant negative impact

```
### Manual plotting:
simple_coeffs <- c(-0.415, 0.142, -0.648, 6.346, -33.916, 9.102, -0.123,
1.091, -0.403, -0.025, -2.157, -0.950)
multiple_coeffs <- c(-0.121, 0.046, 0.013, 2.839, -18.758, 3.658, 0.003,
-1.490, 0.289, -0.012, -0.937, -0.552)

# 2. Plot the points
png("coeff_manual.png", width=800, height=600)
plot(simple_coeffs, multiple_coeffs,
     main="Simple vs. Multiple Regression Coefficients",
     xlab="Univariate (Simple) Coefficients",
     ylab="Multiple Regression Coefficients",
     pch=19, col="blue")

# 3. Add a 45-degree reference line (where x = y)
abline(0, 1, col="red", lty=2)
dev.off()

## Suggested method
# Collect the 12 simple coefficients (slopes)
png("coeff_suggested.png", width=800, height=600)
simple_coeffs <- c(coef(lm.fit_crim)[2], coef(lm.fit_zn)[2],
                   coef(lm.fit_indus)[2], coef(lm.fit_chas)[2],
                   coef(lm.fit_nox)[2], coef(lm.fit_rm)[2],
                   coef(lm.fit_age)[2], coef(lm.fit_dis)[2],
```

```
                    coef(lm.fit_rad)[2], coef(lm.fit_tax)[2],
                    coef(lm.fit_ptratio)[2], coef(lm.fit_lstat)[2])

# Collect the 13 multiple coefficients (excluding the intercept)
multiple_coeffs <- coef(lm.fit_Boston)[-1]

plot(simple_coeffs, multiple_coeffs,
     pch = 19, col = "blue",
     xlab = "Univariate (Simple) Coefficients",
     ylab = "Multiple Regression Coefficients",
     main = "Simple vs. Multiple Regression Coefficients")

# Add a 45-degree line to see which ones changed most
abline(0, 1, col = "red", lty = 2)
dev.off()
```

(d) Check for evidence of non-linear associations. For each predictor X, fit a cubic polynomial model of the form:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon.$$

Addressing individual SLR models and not the MLR model:

Univariate Regression R2 Values

Predictor	Multiple R2	Percentage of Variance Explained
lstat	0.5441	54.41%
rm	0.4835	48.35%
ptratio	0.2578	25.78%
indus	0.2340	23.40%
tax	0.2195	21.95%
nox	0.1826	18.26%
crim	0.1508	15.08%
rad	0.1456	14.56%
age	0.1421	14.21%
zn	0.1299	12.99%
dis	0.06246	6.25%
chas	0.03072	3.07%

Char was ommited as its a dummy, categorical variable. > "The variable `chas` was excluded from the polynomial analysis because it is a qualitative dummy variable. Since it only takes values of 0 and 1, the terms X, X2, and X3 are identical, making it mathematically impossible to fit a polynomial model of degree greater than 1."

```
## Manual method:
### Derived R^2 readings from previous summary functions
### Polynomial models:
lm.fit_polycrim <- lm(medv ~ poly(crim, 3), data = Boston)
lm.fit_polyzn      <- lm(medv ~ poly(zn, 3), data = Boston)
lm.fit_polyindus   <- lm(medv ~ poly(indus, 3), data = Boston)
lm.fit_polynox     <- lm(medv ~ poly(nox, 3), data = Boston)
lm.fit_polyrm      <- lm(medv ~ poly(rm, 3), data = Boston)
lm.fit_polyage     <- lm(medv ~ poly(age, 3), data = Boston)
lm.fit_polydis     <- lm(medv ~ poly(dis, 3), data = Boston)
lm.fit_polyrad     <- lm(medv ~ poly(rad, 3), data = Boston)
lm.fit_polytax     <- lm(medv ~ poly(tax, 3), data = Boston)
lm.fit_polyptratio <- lm(medv ~ poly(ptratio, 3), data = Boston)
lm.fit_polylstat   <- lm(medv ~ poly(lstat, 3), data = Boston)

summary(lm.fit_polycrim)
summary(lm.fit_polycrim)
summary(lm.fit_polyzn)
summary(lm.fit_polyindus)
summary(lm.fit_polynox)
summary(lm.fit_polyrm)
summary(lm.fit_polyage)
summary(lm.fit_polydis)
summary(lm.fit_polyrad)
summary(lm.fit_polytax)
summary(lm.fit_polyptratio)
summary(lm.fit_polylstat)

## Suggested model:
# Get names of all predictors, predictors but specifically exclude 'medv'
AND 'chas'(dummy categorical)
predictors <- names(Boston)[!names(Boston) %in% c("medv", "chas")]

# Fit a cubic model for each predictor and store in a list
cubic_models <- lapply(predictors, function(x) {
  formula <- as.formula(paste("medv ~ poly(", x, ", 3)"))
  return(lm(formula, data = Boston))
})
```

```
# Name the list elements for easy access
names(cubic_models) <- predictors

summary(cubic_models$crim)
summary(cubic_models$zn)
summary(cubic_models$indus)
summary(cubic_models$nox)
summary(cubic_models$rm)
summary(cubic_models$age)
summary(cubic_models$dis)
summary(cubic_models$rad)
summary(cubic_models$tax)
summary(cubic_models$ptratio)
summary(cubic_models$lstat)
```

Results

manual model

Predictor	Multiple R2	P-value (Degree 2)	P-value (Degree 3)	Evidence of Non-Linearity?
crim	0.2177	< 0.001 ***	0.0254 *	**Yes** (Strong Quadratic & Cubic)
zn	0.1649	0.0227 *	< 0.001 ***	**Yes** (Strong Quadratic & Cubic)
indus	0.2768	< 0.001 ***	0.0180 *	**Yes** (Strong Quadratic & Cubic)
nox	0.1939	0.0939	0.0410 *	**Yes** (Cubic only)
rm	0.5612	< 0.001 ***	< 0.001 ***	**Yes** (Strong Quadratic & Cubic)
age	0.1566	0.0061 **	0.3097	**Yes** (Quadratic only)
dis	0.1050	< 0.001 ***	0.0215 *	**Yes** (Strong Quadratic & Cubic)
rad	0.1767	0.0105 *	< 0.001 ***	**Yes** (Strong Quadratic & Cubic)
tax	0.2261	0.0657	0.3531	**No** (Weak/Non-significant)
ptratio	0.2669	0.1090	0.0590	**No** (Weak/Non-significant)
lstat	0.6578	< 0.001 ***	< 0.001 ***	**Yes** (Strong Quadratic & Cubic)

suggested model

Predictor	Multiple R2	Poly Degree 2 (p-value)	Poly Degree 3 (p-value)	Evidence of Non-Linearity?
crim	0.2177	< 0.001 (***)	0.0254 (*)	**Yes** (Strong Cubic)
zn	0.1649	0.0227 (*)	< 0.001 (***)	**Yes** (Strong Cubic)
indus	0.2768	< 0.001 (***)	0.0180 (*)	**Yes** (Strong Cubic)
nox	0.1939	0.0939 (.)	0.0410 (*)	**Yes** (Cubic)
rm	0.5612	< 0.001 (***)	< 0.001 (***)	**Yes** (Strong Cubic)
age	0.1566	0.0061 (**)	0.3097	**Yes** (Quadratic only)
dis	0.1050	< 0.001 (***)	0.0215 (*)	**Yes** (Strong Cubic)
rad	0.1767	0.0105 (*)	< 0.001 (***)	**Yes** (Strong Cubic)
tax	0.2261	0.0657 (.)	0.3531	**No** (Weak Quadratic)
ptratio	0.2669	0.1090	0.0590 (.)	**No** (Weak Cubic)
lstat	0.6578	< 0.001 (***)	< 0.001 (***)	**Yes** (Strong Cubic)